Narration Script

BANTA-Y rationale video v1 · how MCS comes into play · version without narration

Length 2:38 · 1920 × 1080 · 30 fps · subtitles burned in · 301 words in four scenes, then a 10-second title card with no narration. Read each line over **BANTA-Y_Rationale_Video_v1_MCS_No_Narration.mp4**. Each line starts when its subtitle appears, and every visual cue is tied to a word, so match your pace to the subtitles.

How to record

1. Play the video and start speaking when each subtitle appears (**Start**).
2. Finish the line by **Finish by**. **Read time** is the time the line needs at an easy pace.
3. Record each scene as its own take. Scene windows: 0:00–0:21.5, 0:21.5–0:44.8, 0:44.8–1:08.9, 1:08.9–2:27.5, then the title card 2:27.5–2:38 (no narration).
4. The boom lands at 2:08.1. Read the conclusion (4.10 to 4.12) after it, with a full pause before the last line, then stay silent through the title card to 2:38.
5. Pace is about 2.3 words per second, with a short breath at each comma.

Say it this way

MCS: "M-C-S", letter by letter
BANTA-Y: "BAN-tai", like the Filipino word for watch
Aedes aegypti: ay-EE-deez ee-JIP-tie
Albay: al-BYE
Legazpi: leh-GAHS-pee
barangay: ba-rang-GUY
R-nought: "R naught"
CUSUM: KYOO-sum
"test network", never "Albay", for the D1 numbers
Table values and sources are shown on screen only. Do not read them.

Scene 1 · Pupil

0:00.0–0:21.5 · 44 words · bar: Health

#	Start	Finish by	Read time	Narration (read this) · on screen
1.1	0:00.3	0:06.3	6.1 s	**September 2026. A Grade 5 pupil in Legazpi wakes up with a high fever.** On screen: "SEPTEMBER 2026"; a house at dawn; a pupil with a thermometer; the badge "DENGUE" fades in.
1.2	0:06.8	0:11.1	4.3 s	**The clinic writes down one thing first: her home address.** On screen: A case-report card slides in: "Case report · Address: home barangay"; a pin drops on the house.
1.3	0:11.6	0:20.1	8.7 s	**But the mosquito that bit her was not at home. It was at school, in the middle of the day.** On screen: The sun rises; a dotted path runs from the house to the school; an Aedes mosquito circles the school; the house pin greys out.

Scene 2 · Province

0:21.5–0:44.8 · 47 words · bar: Province

#	Start	Finish by	Read time	Narration (read this) · on screen
2.1	0:21.8	0:23.5	1.7 s	**Zoom out to Albay.** On screen: Albay as a field of 720 barangay dots in 18 towns.
2.2	0:24.0	0:31.3	7.4 s	**By mid-September, the province had 2,179 cases and 14 deaths, 84 percent more than the year before.** On screen: Counters: 2,179 cases · 14 deaths · +84% vs the same period of 2025 (Jan 1 to Sept 12, 2026); source Daily Tribune, 2026.
2.3	0:31.8	0:38.2	6.5 s	**The province declared a state of calamity, and schools closed for a week of cleanup.** On screen: Badge "STATE OF CALAMITY"; a school with the sign "Closed Sept 14 to 18 · cleanup and fogging".
2.4	0:38.7	0:43.4	4.8 s	**Sustainable Development Goal 3 asks us to end epidemics like dengue.** On screen: Chips: "SDG 3.3 · end the epidemics of neglected tropical diseases" and "SDG 3.d · early warning".

Scene 3 · Place

0:44.8–1:08.9 · 49 words · bar: Place

#	Start	Finish by	Read time	Narration (read this) · on screen
3.1	0:45.1	0:51.0	6.1 s	**Every case is filed by home address, so the brigades go to the homes.** On screen: Test network D1 (synthetic): bars of cases by residence; B2 has the most; brigade markers move to B2.
3.2	0:51.6	0:56.7	5.2 s	**But Aedes aegypti bites by day, where people gather: classrooms, markets, terminals.** On screen: Daytime arrows flow from every barangay into B1, the poblacion with the central school and public market; B1 glows.
3.3	0:57.2	1:01.9	4.8 s	**Spray where cases live, and you may spray the wrong barangay.** On screen: Spray cloud over B2 while B1 stays red; stamp "WRONG PLACE?".
3.4	1:02.5	1:07.6	5.2 s	**So where is dengue really spreading, and how much control is enough?** On screen: Three lines build: "Where does it spread? · How much is enough? · When do we warn?"

Scene 4 · How MCS comes into play

1:08.9–2:27.5 · 161 words · bar: MCS

#	Start	Finish by	Read time	Narration (read this) · on screen
4.1	1:09.2	1:12.2	3.0 s	**This is where MCS comes into play.** On screen: The letters M, C and S drop in, then unfold into Mathematical, Computational, Sciences.

4.2	1:12.8	1:19.1	6.5 s	**Mathematics writes each barangay's day as a matrix: who spends how many hours, and where.** On screen: Chain lights "Open data → Graph"; panel MATHEMATICS: the 6 × 6 residence-time matrix P of test network D1 fills in as a heat map; $p_{ii} = 1 - \omega_i$, $p_{ij} \propto A_j e^{(-\beta T_{ij})}$.
4.3	1:19.7	1:26.1	6.5 s	**Its largest eigenvalue gives the reproduction number, and dengue fades when it falls below one.** On screen: $M = \kappa D_N P W P^T$, $R0^2 = \lambda_{max}(S)$; dial shows R0 = 1.214 with the threshold line at 1.
4.4	1:26.6	1:32.2	5.7 s	**Computing splits that number into shares by place, and ranks the true hotspots.** On screen: Chain lights "Algorithm → Numbers"; panel COMPUTING: paired bars, cases by residence vs transmission by place; B1 rises to 42.6%.
4.5	1:32.7	1:40.8	8.3 s	**Optimisation finds the cheapest brigade plan that brings the number below one, with a certificate that nothing cheaper exists.** On screen: Chain lights "Optimisation"; panel OPTIMISATION: min $\Sigma c_j u_j$ subject to $R0(u) \le 0.95$; cost bars for five strategies, least cost 22.60 team-days with its certified lower bound.
4.6	1:41.3	1:50.3	9.1 s	**On the test network, cases pointed to a residential barangay, but forty-three percent of transmission was at the school and market.** On screen: Two cards: "Most cases by home: B2, 28.5%" and "Most transmission by place: B1, 42.6%" (test network D1, synthetic).
4.7	1:50.8	1:55.5	4.8 s	**The least-cost plan needed forty percent fewer team-days than spraying everywhere.** On screen: Counter: 37.47 → 22.60 team-days, −40%; versus spraying where cases live 38.88 (−42%).
4.8	1:56.0	2:03.3	7.4 s	**And a CUSUM alarm warned more than a week and a half sooner than a single-week threshold.** On screen: CUSUM curve S_t climbs past h = 12.75; bars of mean delay: CUSUM 4.84 weeks vs single-week rule 6.52 weeks at the same 5% false-alarm rate (5,000 simulated years).
4.9	2:03.8	2:07.2	3.5 s	**Twelve results, proven, then verified against independent code.** On screen: Chain lights "Proof"; twelve proof squares turn into green checks; 300/300 reproduction numbers, 120/120 control plans, 10,000 alarm sequences.
4.10	2:08.4	2:10.9	2.6 s	**BANTA-Y will not kill one mosquito.** On screen: Boom. Screen darkens; the line appears in white.
4.11	2:11.5	2:18.3	7.0 s	**But it will tell the brigades where the bites really happen, and how much is enough.** On screen: The line appears in gold.

4.12	2:19.7	2:25.2	5.7 s	**[pause] Homes tell you where dengue is reported. Places tell you where it spreads.** On screen: Two lines: "Homes tell you where dengue is reported." (white), then "Places tell you where it spreads." (gold).

Title card · Researchers

2:27.5–2:38 · no narration

#	Start	Finish by	Read time	Narration (read this) · on screen
5.1	2:27.5	2:38	—	**[No narration. Stay silent; let the music and the title card play.]** On screen: the school building under a deep red overlay, the school seal top-left and a white corner grid; the gold letters BANTA - Y drop in one by one with a hit, then a gold shine; the full paper title in white; two thin white lines draw outward; "De Asis, Dave Justin Ll." in large gold with "LEAD RESEARCHER · FOUNDER OF THE BANTA-Y RESEARCH & APPLICATION"; then "Brimbuela, Peter John A. · Shin, Yeaseong" in white. Fade to black at 2:38.



Title card, final frame (also supplied as BANTA-Y_Title_Page_1920×1080.png and BANTA-Y_Title_Page_3840×2160.png).

Clean read

The same narration without the production notes. Use it as a teleprompter or to rehearse. A slash marks a short breath.

SCENE 1 · PUPIL · 0:00.0–0:21.5

0:00.3 September 2026. A Grade 5 pupil in Legazpi wakes up with a high fever.

0:06.8 The clinic writes down one thing first: her home address.

0:11.6 But the mosquito that bit her was not at home. It was at school, / in the middle of the day.

SCENE 2 · PROVINCE · 0:21.5–0:44.8

0:21.8 Zoom out to Albay.

0:24.0 By mid-September, / the province had 2,179 cases and 14 deaths, / 84 percent more than the year before.

0:31.8 The province declared a state of calamity, / and schools closed for a week of cleanup.

0:38.7 Sustainable Development Goal 3 asks us to end epidemics like dengue.

SCENE 3 · PLACE · 0:44.8–1:08.9

0:45.1 Every case is filed by home address, / so the brigades go to the homes.

0:51.6 But Aedes aegypti bites by day, / where people gather: classrooms, / markets, / terminals.

0:57.2 Spray where cases live, / and you may spray the wrong barangay.

1:02.5 So where is dengue really spreading, / and how much control is enough?

SCENE 4 · HOW MCS COMES INTO PLAY · 1:08.9–2:27.5

1:09.2 This is where MCS comes into play.

1:12.8 Mathematics writes each barangay's day as a matrix: who spends how many hours, / and where.

1:19.7 Its largest eigenvalue gives the reproduction number, / and dengue fades when it falls below one.

1:26.6 Computing splits that number into shares by place, / and ranks the true hotspots.

1:32.7 Optimisation finds the cheapest brigade plan that brings the number below one, / with a certificate that nothing cheaper exists.

1:41.3 On the test network, / cases pointed to a residential barangay, / but forty-three percent of transmission was at the school and market.

1:50.8 The least-cost plan needed forty percent fewer team-days than spraying everywhere.

1:56.0 And a CUSUM alarm warned more than a week and a half sooner than a single-week threshold.

2:03.8 Twelve results, / proven, / then verified against independent code.

2:08.4 BANTA-Y will not kill one mosquito.

2:11.5 But it will tell the brigades where the bites really happen, / and how much is enough.

2:19.7 [pause] Homes tell you where dengue is reported. Places tell you where it spreads.

TITLE CARD · RESEARCHERS · 2:27.5–2:38

[Silent. Title card: BANTA-Y · De Asis, Dave Justin Ll., Lead Researcher · Founder · Brimbuela, Peter John A. · Shin, Yeaseong. Ends 2:38.]